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Project Title:

**DESIGN AND CONFIGURATION OF A SECURED
HIERARCHICAL NETWORK ARCHITECTURE.**

OBJECTIVE:

To build a resilient and secure enterprise network simulation using the 3-tier (Core-Distribution-Access) hierarchical model, focusing on redundancy, fail-over, and protocol implementation

METHODOLOGY:

1. *Design:* Plan the topology, IP scheme, VLANs for a multi-site WAN/MAN scenario.

2. *Build & Configure:* Use *Cisco Packet Tracer* to:

- Set up devices and base IP addressing.
- Implement WAN links (PPP with CHAP security).
- Enable dynamic routing (OSPF) for full connectivity.
- Configure Layer 2 stability (STP/RSTP) and gateway redundancy (HSRP).
- Apply access-layer security (Port Security, VLANs).

3. *Test & Validate:*

Rigorously test end-to-end connectivity and demonstrate failover by simulating link/device failures.

4. Document:

Compile a report with diagrams, configurations, and test results.

KEY TOOLS & PROTOCOLS:

Tool: Cisco Packet Tracer.

Protocols:

1. OSPF (routing)
2. PPP/CHAP (WAN)
3. STP (loop prevention)
4. HSRP (gateway redundancy)
5. VLANs (segmentation).

FINAL DELIVERABLES:

1. A complete project report with design justifications and evidence.
2. A fully functional Packet Tracer (.pkt) file.
3. Device configuration texts and summary presentation.

Success Measured By: A working simulated network where connectivity persists automatically after simulated failures, proving high availability and security as per the design objectives.

This structured approach ensures systematic progress from planning to validation, providing clear milestones for review.